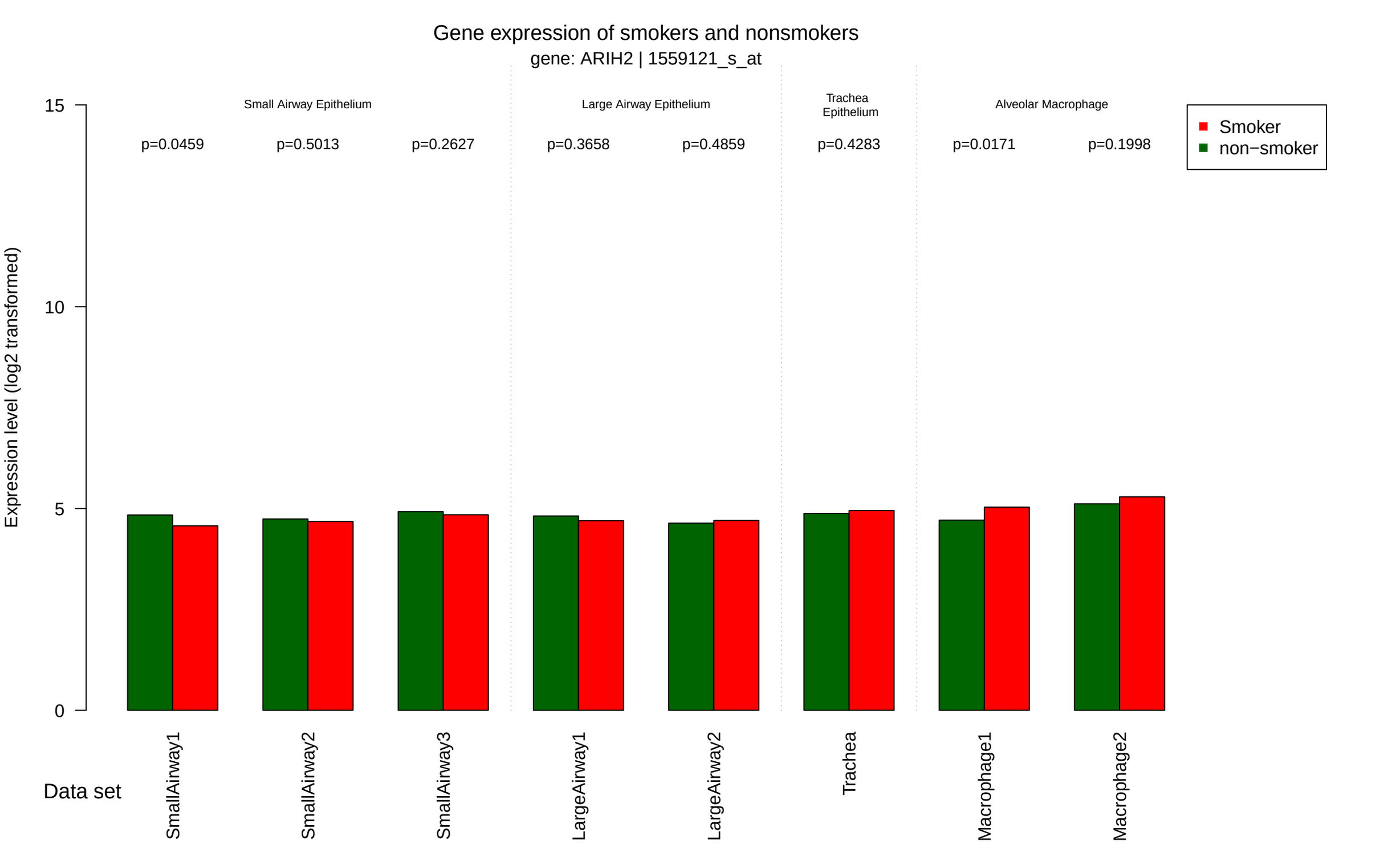
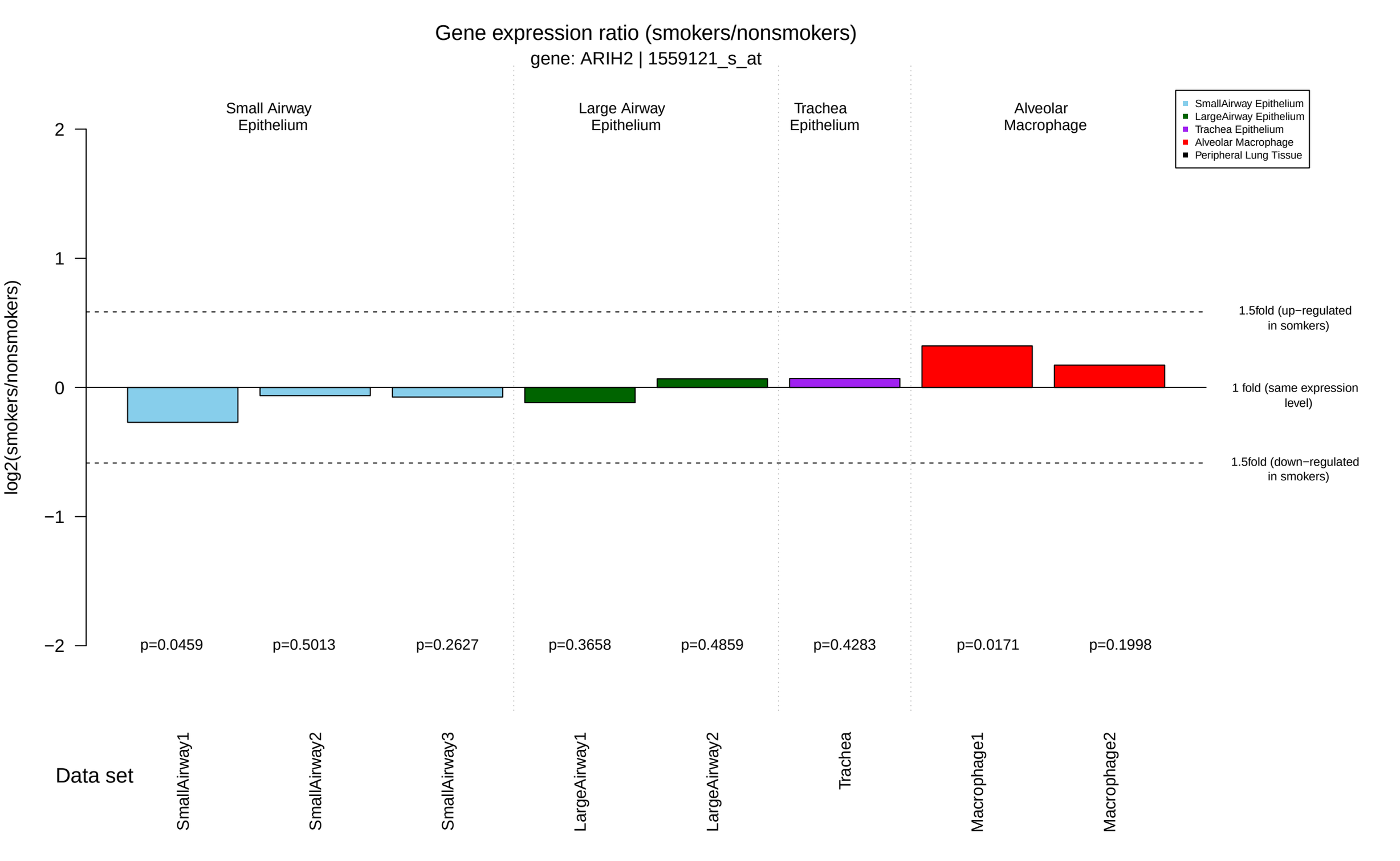
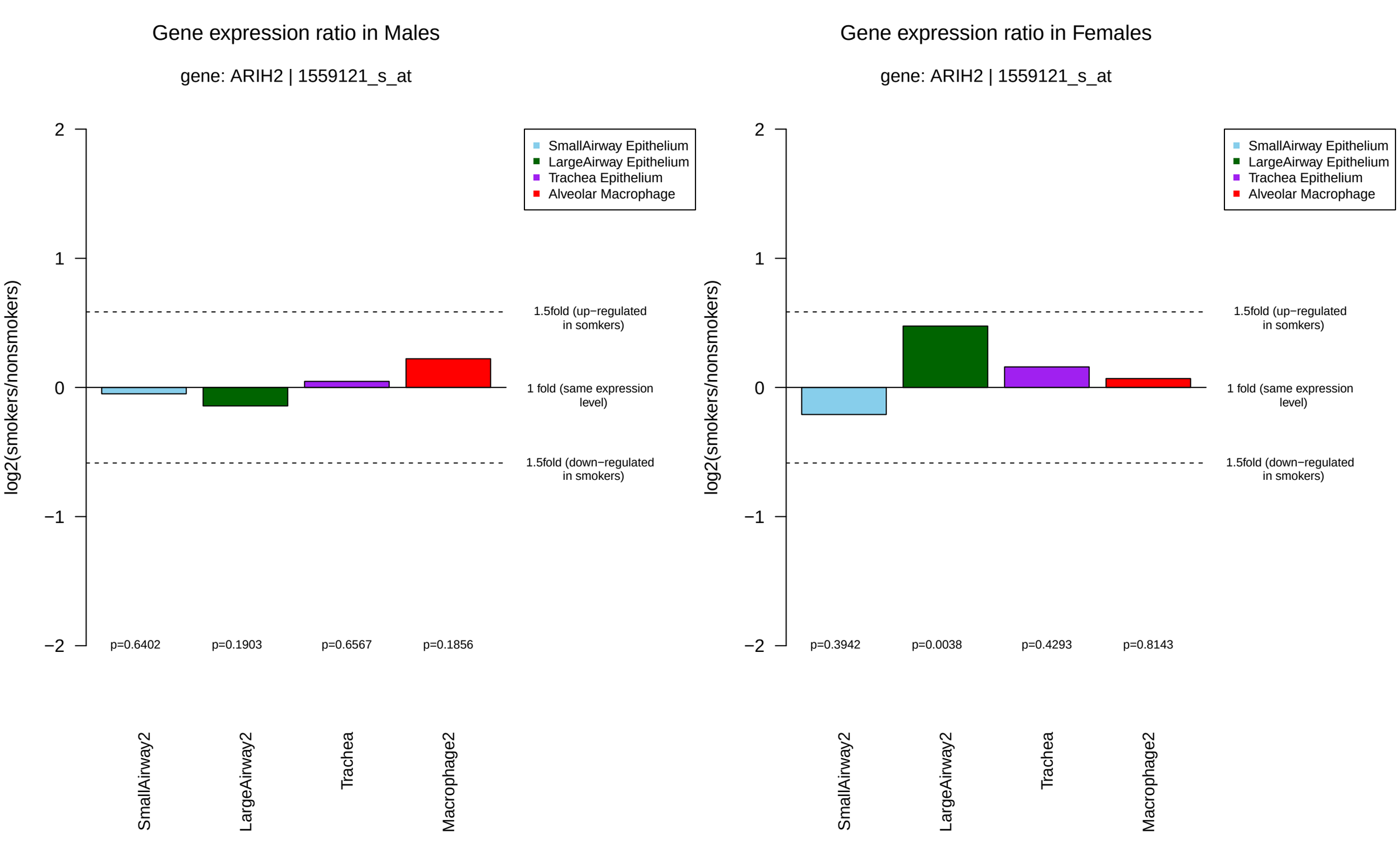
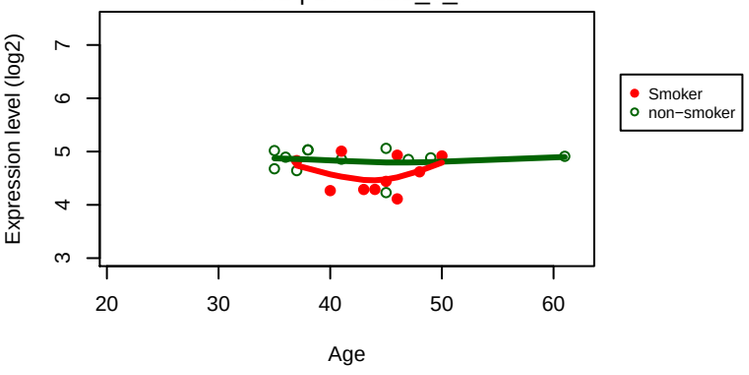
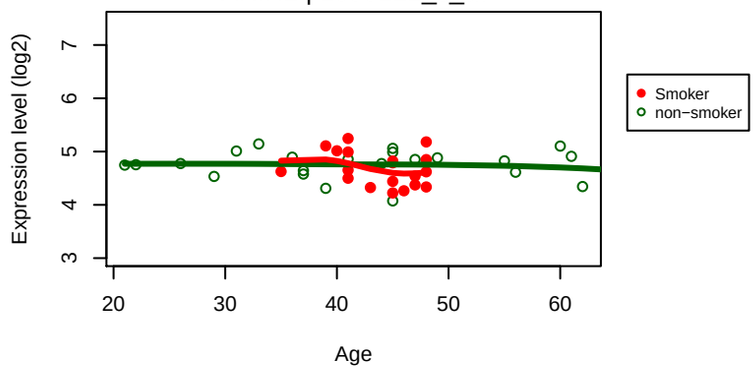
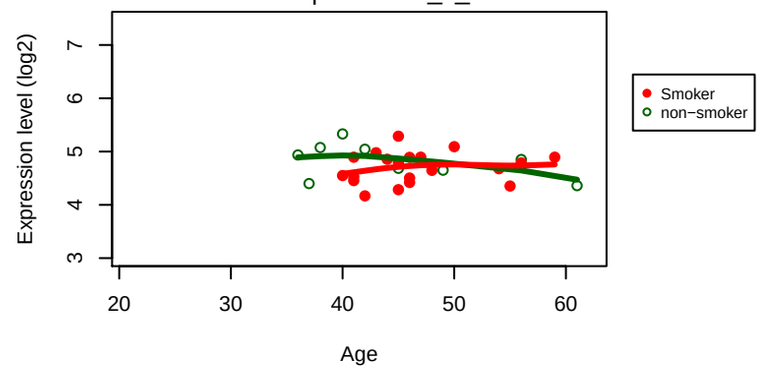

Gene expression of smokers and nonsmokers
gene: ARIH2 | 1559121_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0459
p=0.5013
p=0.2627
p=0.3658
p=0.4859
p=0.4283
p=0.0171
p=0.1998
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

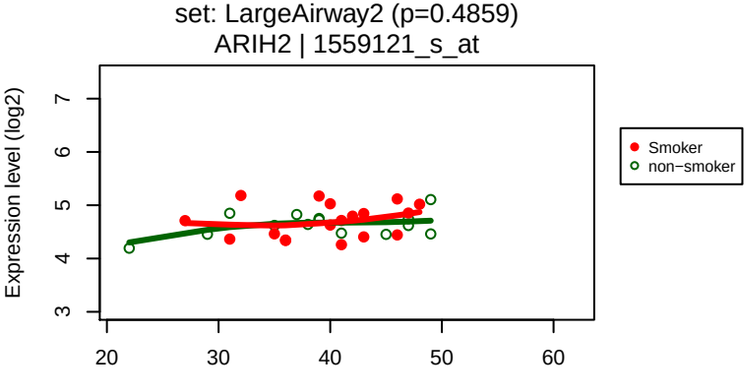

Gene expression ratio (smokers/nonsmokers)
gene: ARIH2 | 1559121_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0459
p=0.5013
p=0.2627
p=0.3658
p=0.4859
p=0.4283
p=0.0171
p=0.1998
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

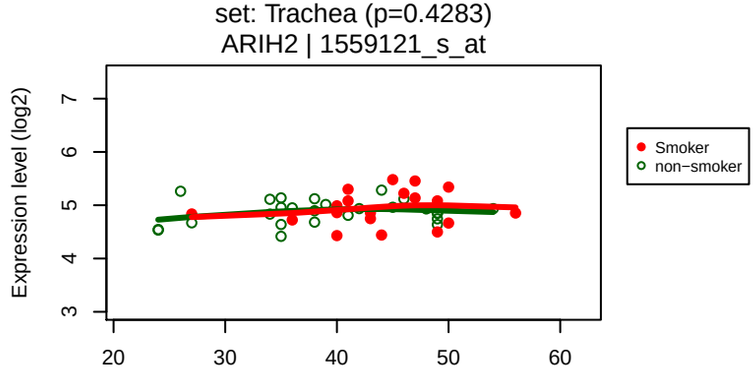

Gene expression ratio in Males
gene: ARIH2 | 1559121_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6402
p=0.1903
p=0.6567
p=0.1856
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARIH2 | 1559121_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3942
p=0.0038
p=0.4293
p=0.8143
SmallAirway2
LargeAirway2
Trachea
Macrophage2

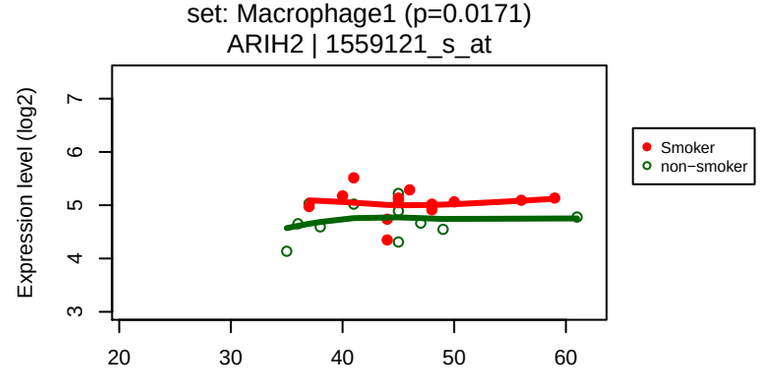

set: SmallAirway1 (p=0.0459)
ARIH2 | 1559121_s_at
Expression level (log2)
Age
Smoker
non−smoker

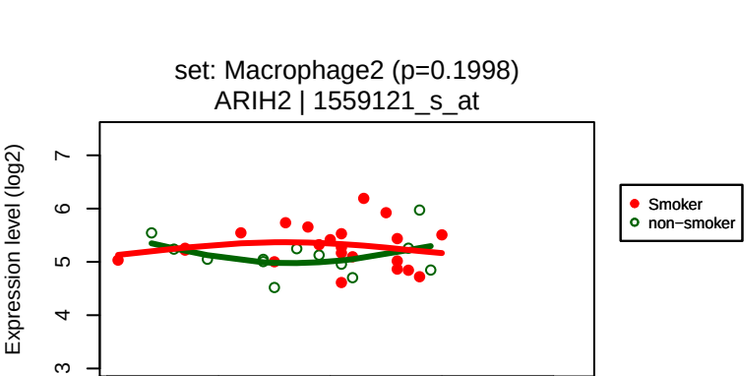

set: SmallAirway2 (p=0.5013)
ARIH2 | 1559121_s_at
Expression level (log2)
Age
Smoker
non−smoker

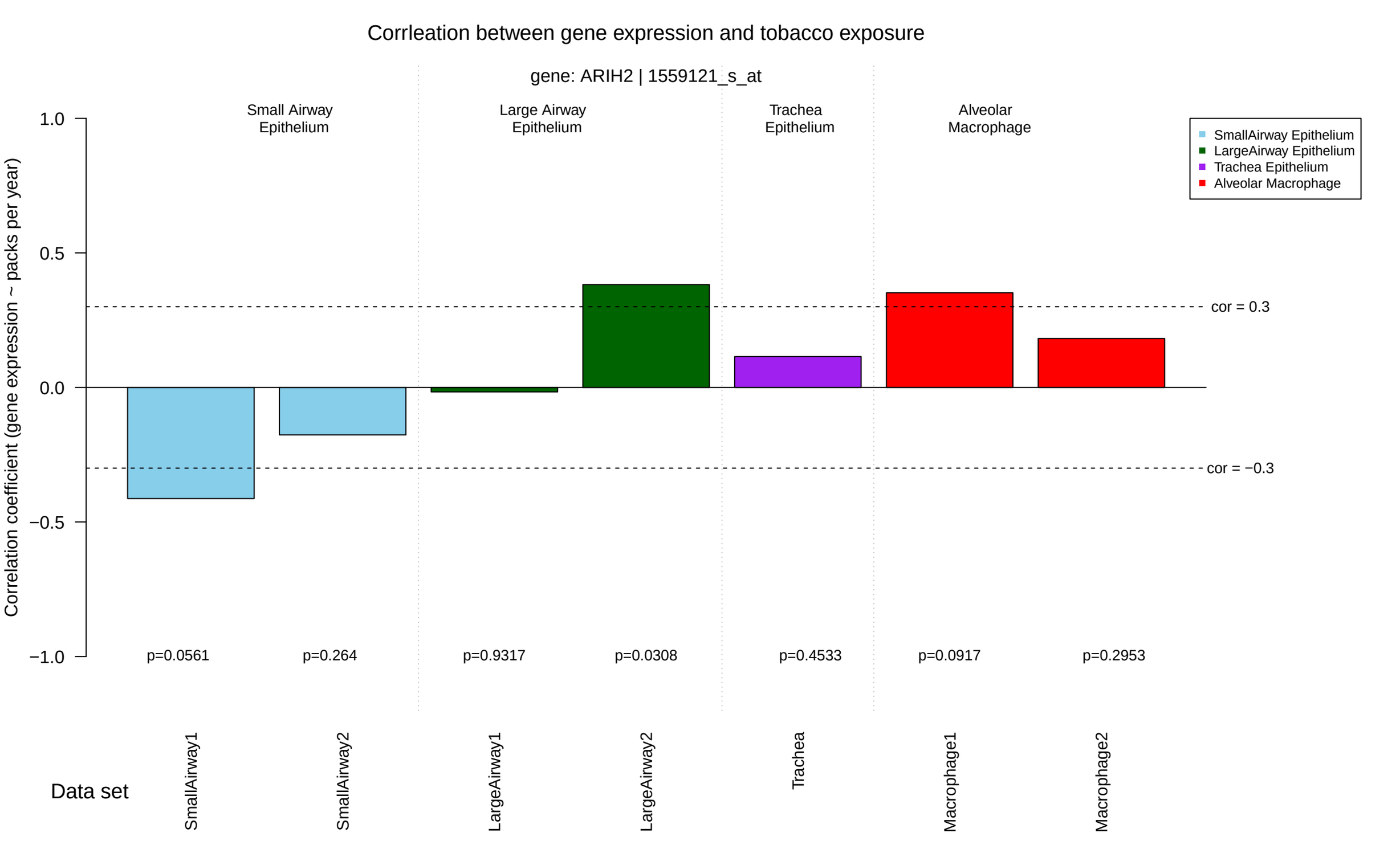

set: LargeAirway1 (p=0.3658)
ARIH2 | 1559121_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4859)
ARIH2 | 1559121_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4283)
ARIH2 | 1559121_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0171)
ARIH2 | 1559121_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1998)
ARIH2 | 1559121_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARIH2 | 1559121_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0561
p=0.264
p=0.9317
p=0.0308
p=0.4533
p=0.0917
p=0.2953
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2